SMARTBRIDGE / SKILL WALLET

Solution Architecture Report

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Team ID	SWTID-2026-6274
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Solution Architecture Diagram

The following architectural representation maps out the Presentation Layer, Application Layer, and Data Layer for automating standard laptop orders in ServiceNow Flow Designer.

Presentation / Client Layer[ServiceNow Service Portal, Employee Self-Service UI, IT Admin Dashboard]
API Gateway / Load Balancer[ServiceNow Instance Router & REST / SOAP Endpoint Gateway]
Auth Service & Core Logic Service[ServiceNow ACLs, System User Roles, Flow Designer Automation Engine]
External API Integrations[Hardware Catalog Integration, Notification & Email Spoke Services]
Data / Storage Layer[ServiceNow MySQL Relational Database (sc_req_item, sc_task, sys_user tables)]

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides user interface for employees to request standard laptops and for IT admins to	ServiceNow Service Portal, HTML5, CSS3, Client Scripts

	track fulfillment.	
[API Gateway]	Manages routing, inbound HTTP requests, and webhooks for automated trigger events.	ServiceNow Application Gateway, REST Endpoints
[Microservices]	Executes automated procurement logic, workflow triggers, task generation, and assignment routing.	ServiceNow Flow Designer, JavaScript, Business Rules
[Database]	Stores catalog items, requested items (RITM), catalog tasks (SCTASK), and user approval states securely.	ServiceNow MySQL Relational Database